

Basic Principles of Medicine 1
Module: Musculoskeletal System | Lecture 27

Histology of the Skin

Dr. Deon Forrester

dforrester@sgu.edu

*Department of Anatomical Sciences
School of Medicine, St. George's University*

Copyright

All year-1 course materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed, or published in any form, printed, electronic, or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

'View only' files, such as lecture recordings, are explicitly excluded from download, and creating copies of these recordings by students and other users is strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St. George's University. Users of this document identifying potential violations of these regulations are asked to bring their concerns to the attention of the author.

Recommended Reading

Histology – A Text and Atlas

Pawlina 9th Edition

Chapter 15: Integumentary System Pg. 524-565

[https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences-
lwwhealthlibrary-com.sgu.idm.oclc.org/book.aspx?bookid=3290](https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences-
lwwhealthlibrary-com.sgu.idm.oclc.org/book.aspx?bookid=3290)

Images from:

- Pawlina 9th Edition
- Wheater's Functional Histology 5th Edition
- Wheater's Basic Pathology 5th Edition
- Virtual microscopy database by the American Association of Anatomists
- Junqueira's basic histology Text and atlas – <https://accessmedicine-mhmedical-com.periodicals.sgu.edu/book.aspx?bookid=1687>
- <http://www.histology.be/atlas/HSH/index.html>

Objectives

Describe the 5 layers of the epidermis and its associated cell types.	SOM.MK.I.BPM1.2.MSK.3.HCB.1054
Describe cytokeratins.	SOM.MK.I.BPM1.2.MSK.3.HCB.1055
Describe keratohyalin granules.	SOM.MK.I.BPM1.2.MSK.3.HCB.1056
Describe the steps in epidermal cell differentiation and cell renewal in relation to the epidermal layers.	SOM.MK.I.BPM1.2.MSK.3.HCB.1057
Describe the steps in the process of keratinization.	SOM.MK.I.BPM1.2.MSK.3.HCB.1058
State the function of the epidermal water barrier.	SOM.MK.I.BPM1.2.MSK.3.HCB.1059
Describe the microanatomical structure of the epidermal water barrier.	SOM.MK.I.BPM1.2.MSK.3.HCB.1060
Compare and contrast the typical epidermal turnover time vs. the hyperproliferative disease psoriasis.	SOM.MK.I.BPM1.2.MSK.3.HCB.1061
List at least 3 factors associated with skin pigmentation.	SOM.MK.I.BPM1.2.MSK.3.HCB.1062
Differentiate between thick and thin skin based on location and histological features.	SOM.MK.I.BPM1.2.MSK.3.HCB.1063

Overview

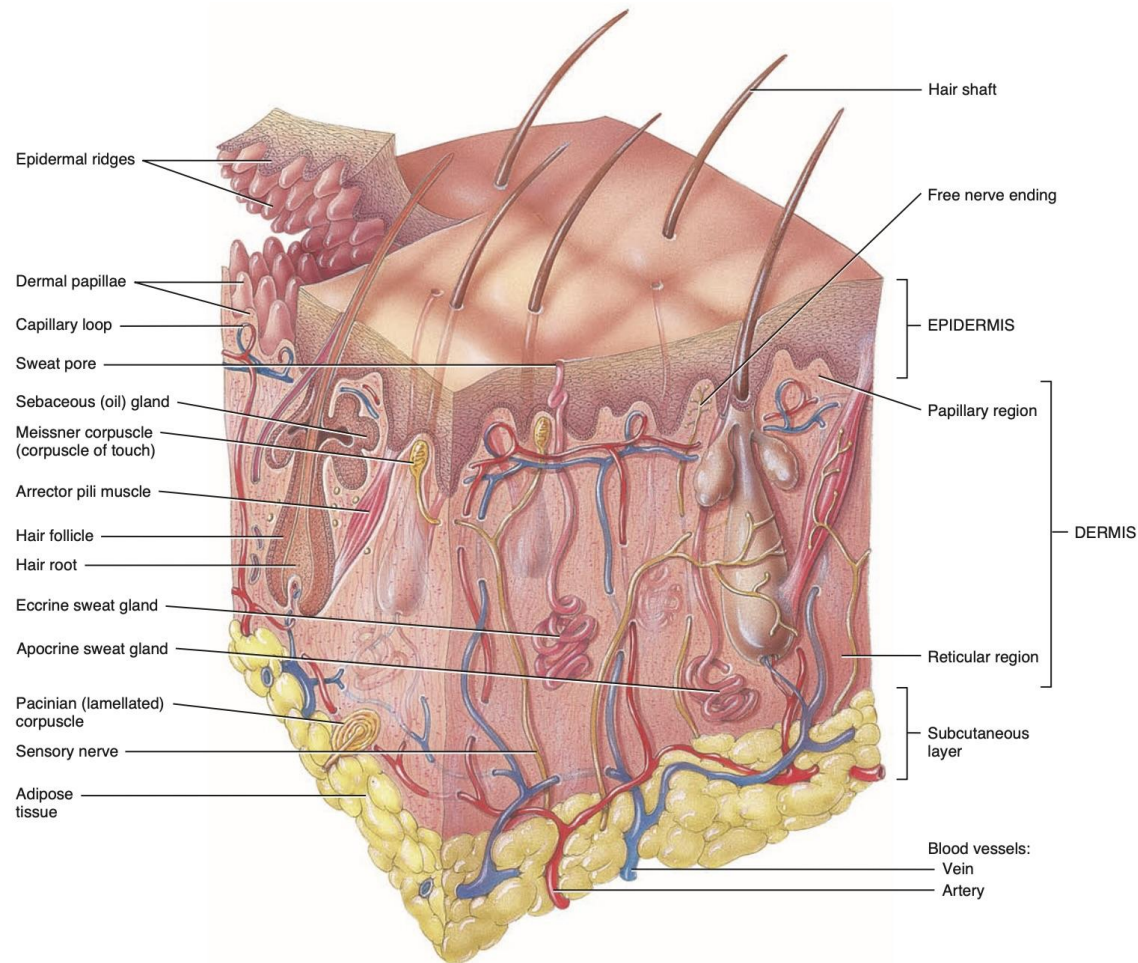
The Integumentary System

- Layers of the skin
 - **Epidermis**
 - **Dermis**
- Hypodermis
- Associated Appendages
- Functions of the skin
- Cells of the skin
 - **Keratinocytes**
 - **Merkel cells**
 - **Melanocytes**
 - **Langerhans' cells**
- Thick skin vs. thin skin
- Epidermis
 - **Stratum basale**
 - **Stratum spinosum**
 - **Stratum granulosum**
 - **Stratum lucidum**
 - **Stratum corneum**
- Epidermal water barrier
- Keratinization
- Epithelial turnover

Integumentary System

Cutis (skin) and its associated appendages

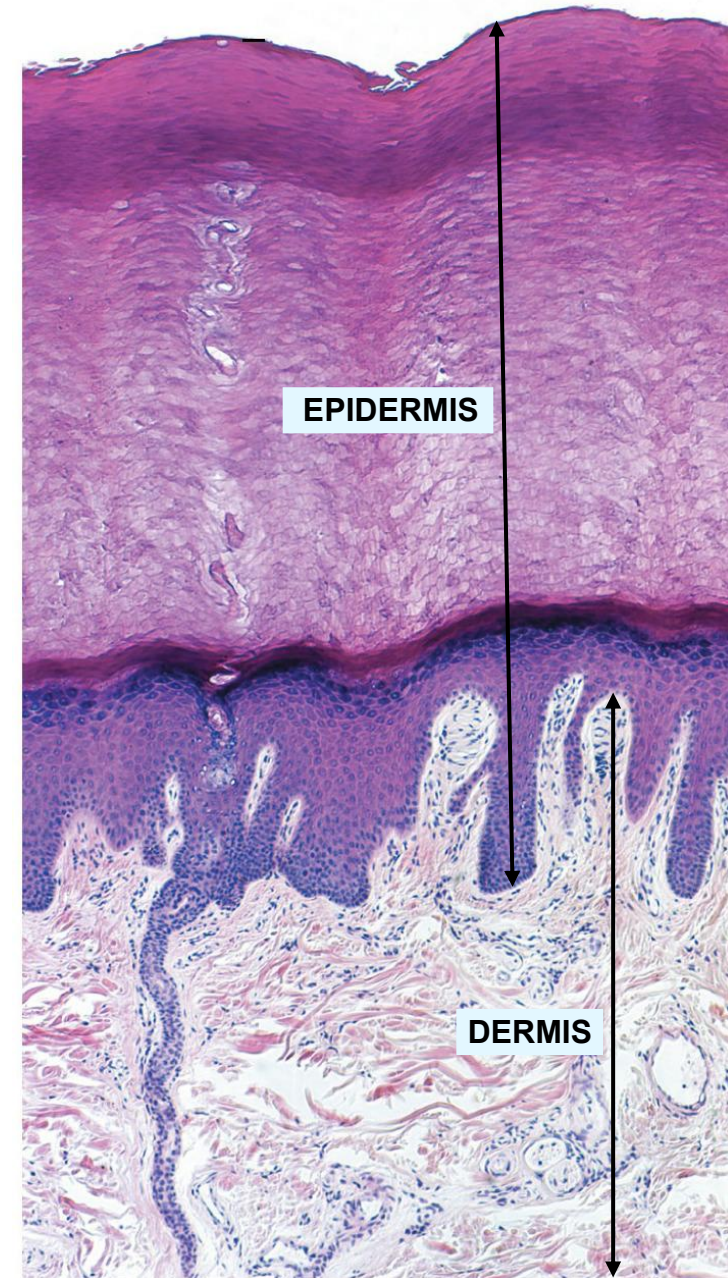
- The largest organ of the human body



Layers of the Skin

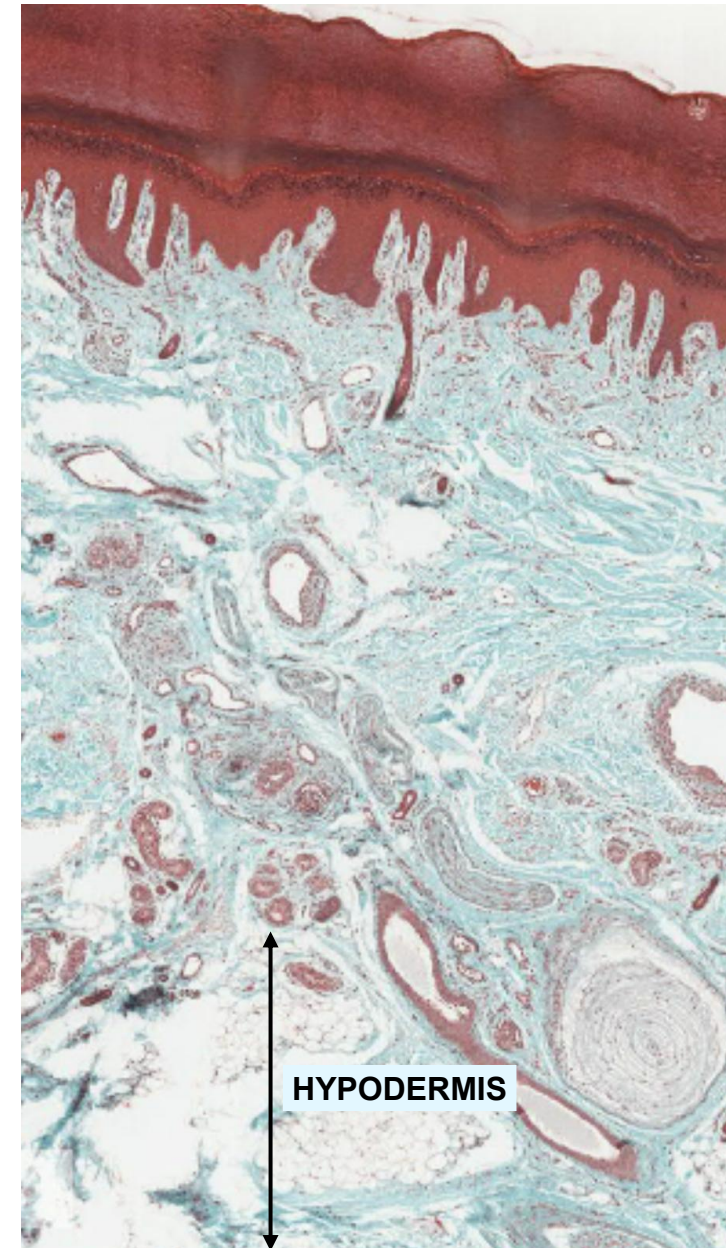
Composed of 2 major layers:

- **Epidermis** – Keratinized stratified squamous epithelium
 - Derived from **ectoderm**
- **Dermis** – Connective tissue layers
 - Derived from **mesoderm**
 - **Papillary layer** – Loose connective tissue
 - **Reticular layer** – Dense irregular connective tissue



Hypodermis

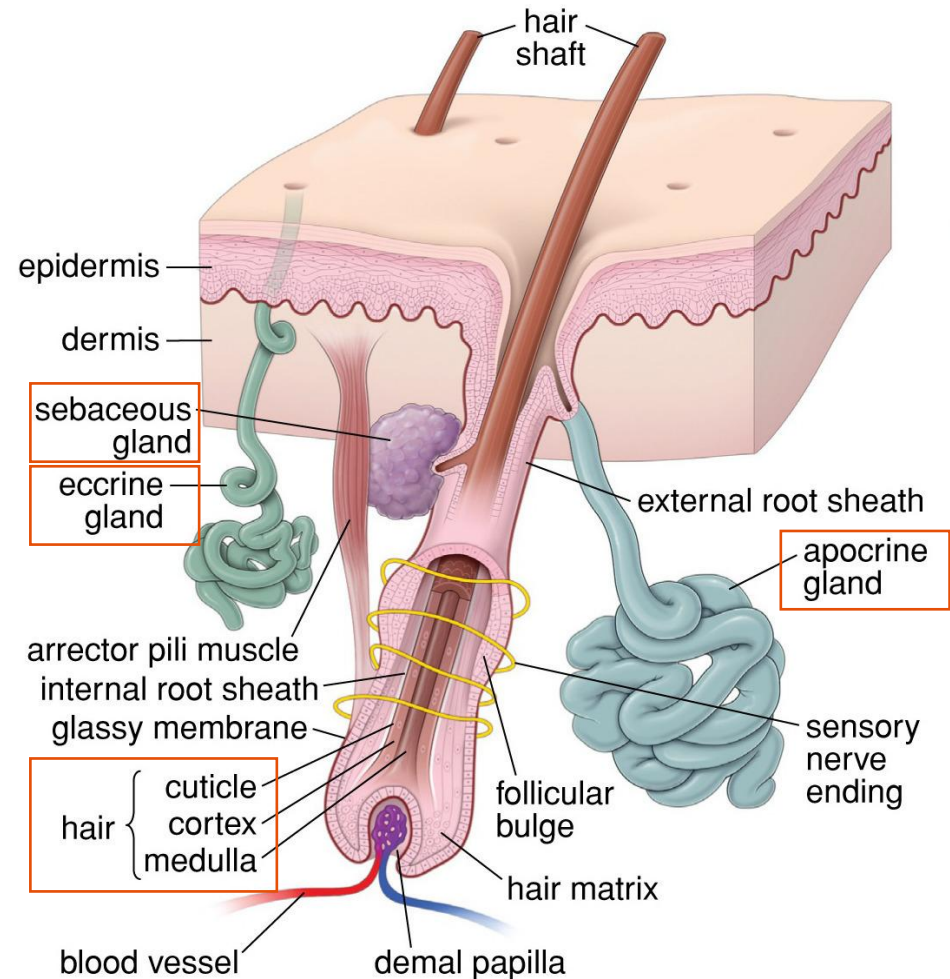
- Located deep to the dermis
 - **NOT** a layer of skin
- Contains variable amounts of adipose tissue (**panniculus adiposus**) arranged in lobules separated by connective tissue septa
 - Major energy storage site
 - Provides insulation
- Associated appendages such as hair follicles, mechanoreceptors and glands can be found in this layer
- Often referred to as **subcutaneous fascia** in gross anatomy
 - Can be very thick in residents of cold countries



Associated Appendages

Epithelial derivatives of the epidermis:

- Hair & hair follicle
- Nails
- Sweat glands
 - Eccrine
 - Apocrine
- Sebaceous glands
- Mammary glands

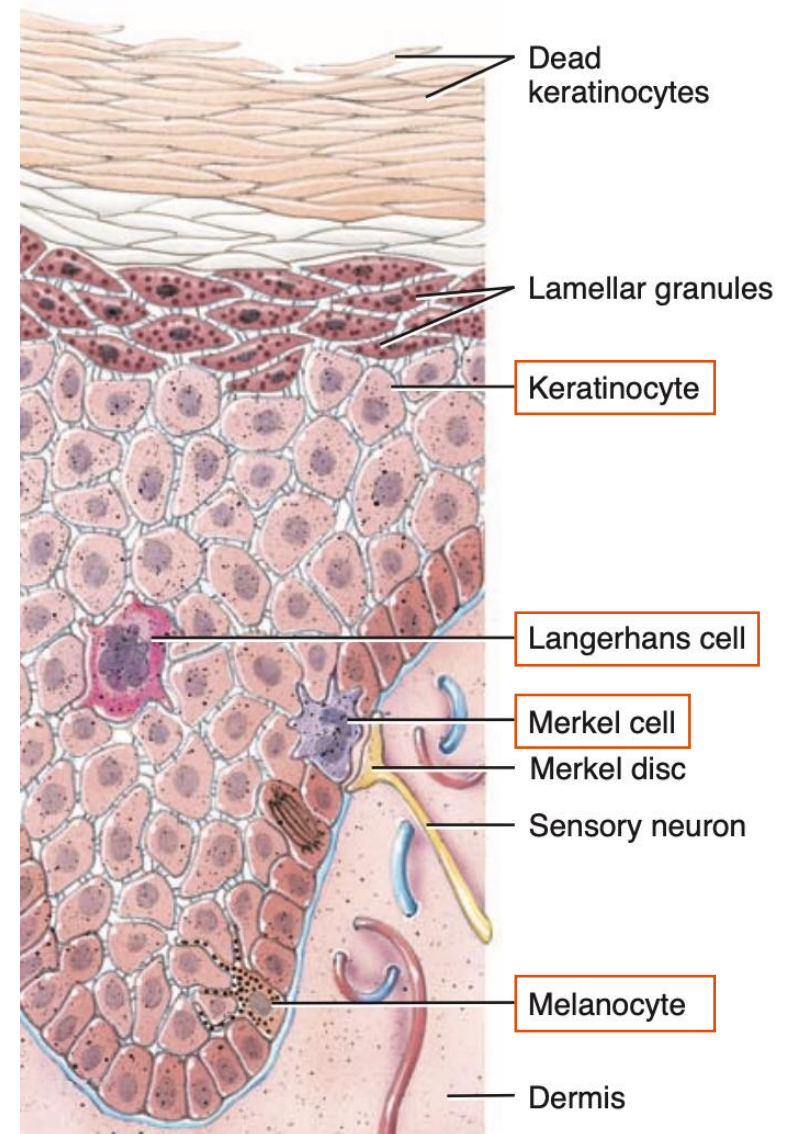


Functions of the Integumentary System

- **Barrier** – protects against physical, chemical, and biological agents in the external environment
- **Immunological role** – processes and presents antigens to cells within lymphatic tissues
- **Homeostasis** – regulates of body temperature and water loss
- **Sensory information** – samples the external environment and relays this information to the central nervous system
- **Endocrine functions** – secretes hormones, cytokines and growth factors; converts vitamin D into its hormonally active form D3
- **Excretion** – secretes sweat and other waste products from the body

Cells of the Integumentary System

- **Keratinocytes** – highly specialized epithelial cells
 - Accounts for **85%** of the cells in the epidermis
- **Merkel cells** – mechanoreceptors
 - Associated with sensory nerve endings
 - Constitutes **6%-10%** of cells in the epidermis
- **Melanocytes** – pigment producing cells of the epidermis
 - Comprises **~5%** of the cell population
- **Langerhans cells** – antigen presenting cells (APC's)
 - Plays a role in the immunological function of skin
 - Accounts for **2%-5%** of cells

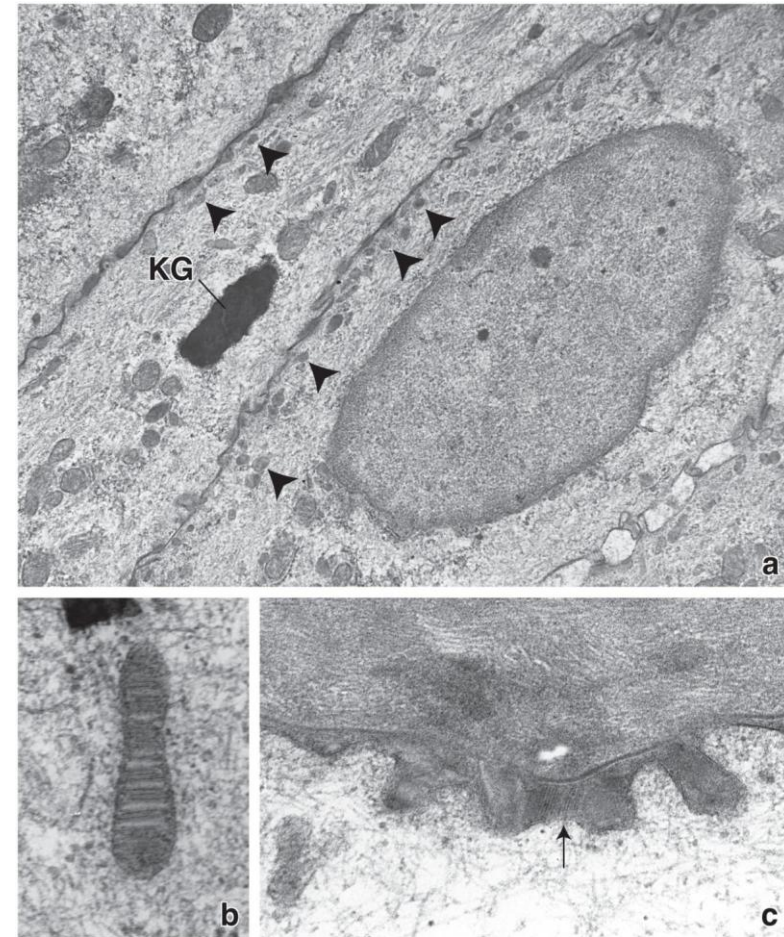


Keratinocytes

- Largest cell population of the epidermis
- Differentiate from the stratum basale, and are present throughout all layers of the epidermis

Functions:

- Produce **keratin filaments** – mainly the keratinocytes of stratum basale
- Produce **keratohyalin granules** – cells in spinosum and granulosum layers
- Produce **lamellar bodies** – cells in spinosum and granulosum layers
- Contribute to the formation of the **epidermal water barrier**



Keratinocytes

Features	Location Within the Epidermis		
	Basale	Spinosum	Granulosum
General	- Cuboidal in appearance - Basophilic staining	- Large cells - Vast cytoplasmic processes - Eosinophilic staining	- Squamous in appearance - Cells begin to keratinize - Basophilic staining
Nucleus	Large nuclei with prominent nucleolus	Ovoid nuclei	Flattened ill-defined nuclei
Organelles	- Perinuclear zone → rich in free ribosomes and mitochondria - Well-developed rER - Small Golgi complex		
Cytoskeletal Element	Keratin filaments present toward the periphery	Abundant keratin filaments	- Keratohyalin granules - Lamellar bodies
Junctional Complex	- Hemidesmosomes → links keratinocytes to BM - Desmosome → links keratinocytes to each other and cells of spinosum layer	Desmosomes → attach neighboring cells to each other	
Other			Ovoid bodies (keratinosomes or Odland bodies) → its contents contains hydrophobic, glycolipid properties that contribute to the water barrier

Merkel cells

- Located in the **stratum basale**
- Possess antigenic markers of both epidermal and neural type
 - Appear **clear** with **lobed nuclei** and **dendritic processes**
 - Cells are adjoined to neighboring keratinocytes via **desmosomes**
 - Cytoplasm contains **neurosecretory granules** and keratin filaments
- Closely associated with the expanded terminal bulb of afferent myelinated nerve fibers – **Merkel's corpuscle**
 - Most abundant in skin where sensory perception is acute (fingertips)

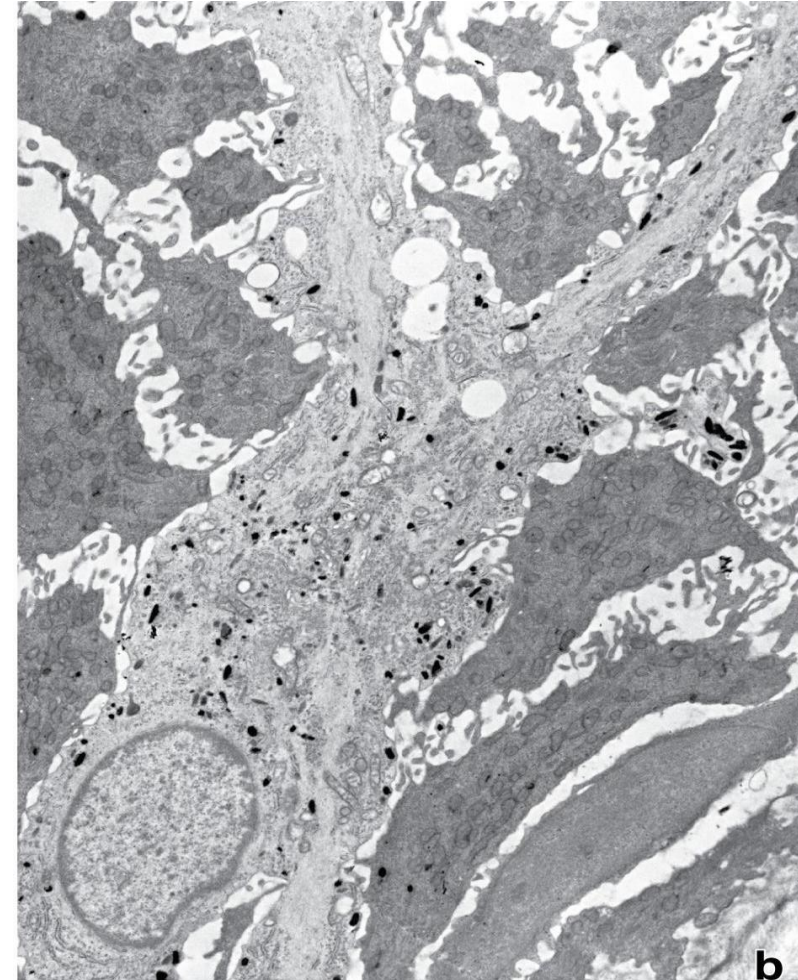
Functions:

- Detect **touch sensations**



Melanocytes

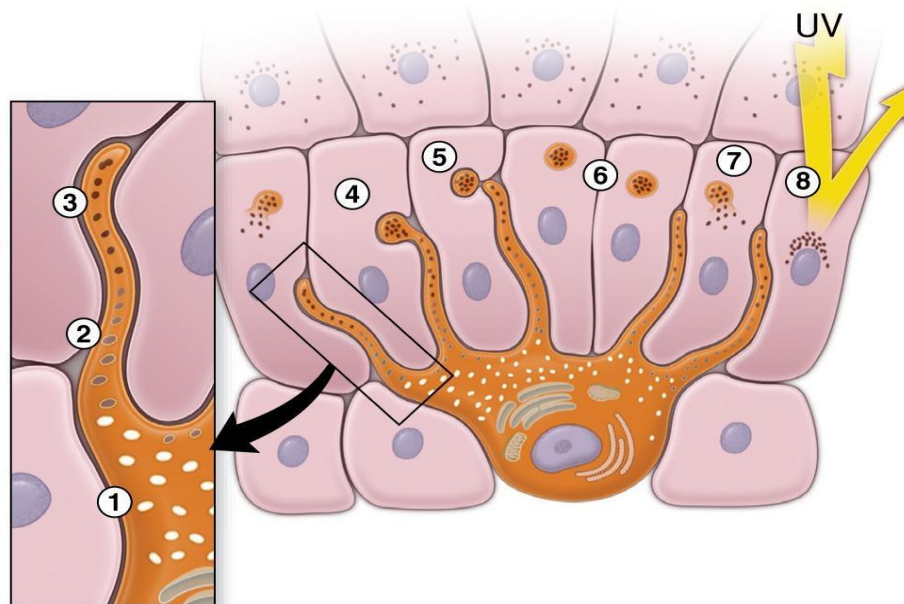
- Located in the **stratum basale**
- Originate from neural crest cells
 - **Clear** cell with **large elongated nuclei** and **multiple cytoplasmic processes** that extend between keratinocytes
 - One melanocyte can communicate with up to 40 keratinocytes
 - Lack desmosome connections but cells attach to the basement membrane via **hemidesmosome-like structures**
 - Cytoplasm contains oval dark staining **premelanosomes** and **melanosomes** (responsible for the synthesis of melanin)
 - Lacks keratin filaments
- Melanocyte to keratinocyte ratio 1:4 to 1:10
 - Varies based on body regions



Melanocytes

Functions:

- **UV protection** – melanin is produced and transferred to surrounding keratinocytes (**cytokrine secretion**)
 - Oxidation of tyrosine to 3,4-dihydroxyphenylalanine (DOPA) and subsequent transformation of DOPA into melanin
- **Skin color** – determined by the amount and ratio of melanin types present
 - Lighter skinned individuals have increased melanin degradation



Factors Associated With Skin Pigmentation

- **UV Radiation** – melanin production is influenced by the surrounding keratinocytes in response to UV radiation
- **Melanin distribution** – dispersed differently in light vs. dark skinned persons
 - Light skinned persons → concentrated in stratum basale
 - Dark skinned persons → dispersed throughout stratum basale and spinosum
- **Melanin type** – each type of melanin has a different function and is driven by different genetic factors
 - The pigment produced by melanocytes consist of two different forms:
 - **Eumelanin**
 - **Pheomelanin**

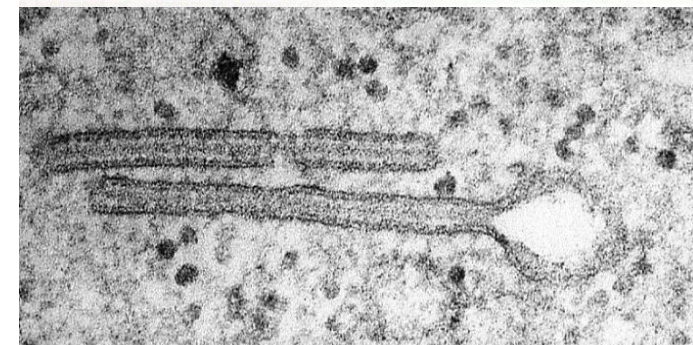
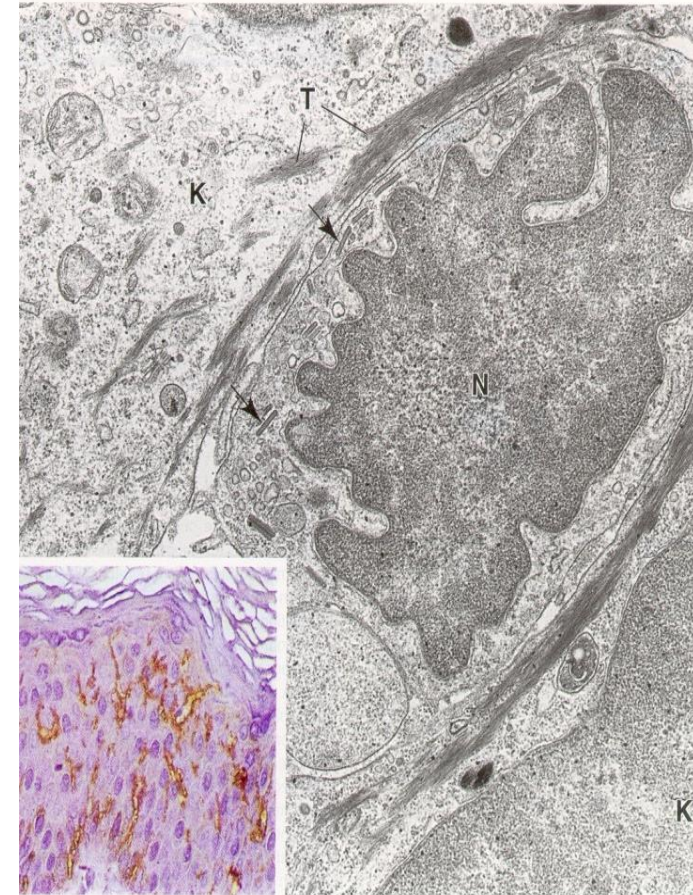
Factors Associated With Skin Pigmentation

- **Eumelanin** and **pheomelanin** have different functions and are driven by different genetic factors
 - The ratio between the two determines skin and hair color

Eumelanin	Pheomelanin
Protection against UVR damage	Allows a greater amount of light to pass through thereby increasing vitamin D production at altitudes where light intensity is low
Scatter and absorb UV rays and eliminate UV-generated free radicals	Found more commonly in atypical nevi and melanomas
Degrades faster in lighter skinned individuals	Increased oxidative damage

Langerhans cells

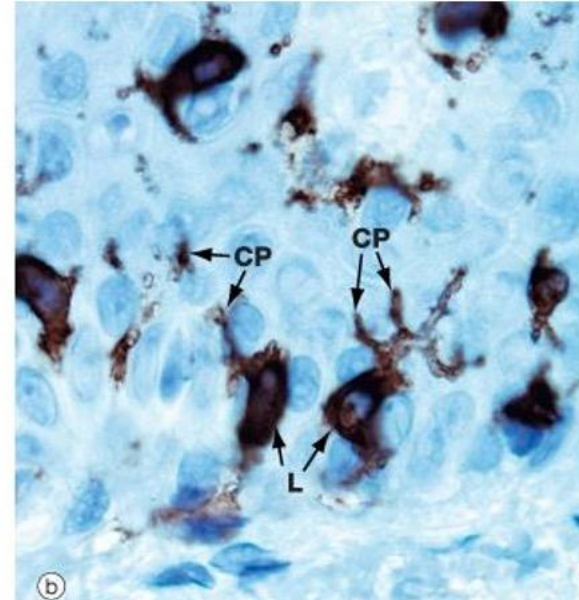
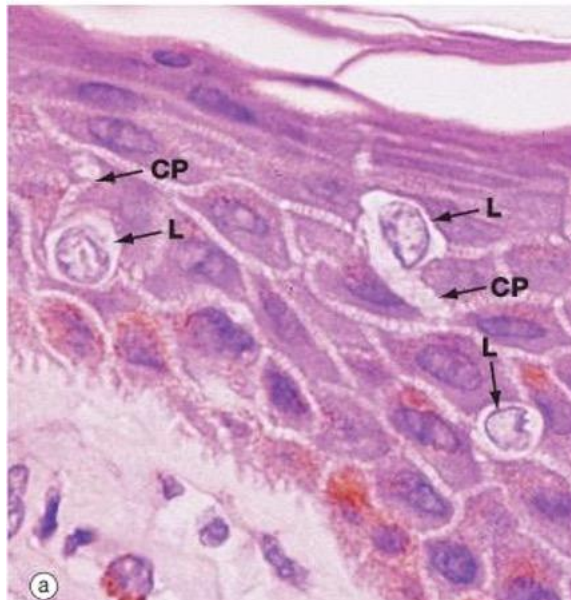
- Least numerous cells in the epidermis
- Present in all epidermal layers but most **easily recognized in stratum spinosum**
- Originates from erythro-myeloid progenitor
 - **Clear** cytoplasm with an **indented nuclear membrane** (uneven profile) and **dendritic processes**
 - Lack desmosome connections with neighboring keratinocytes
 - Cytoplasm contains **Birbeck granules** – tennis racquet shaped granules
 - Small linear (rod-like) structure with a bulbous expansion (vesicle) at one end
 - Regular cross striations
 - Function unclear but may be part of the endosome system



Langerhans cells

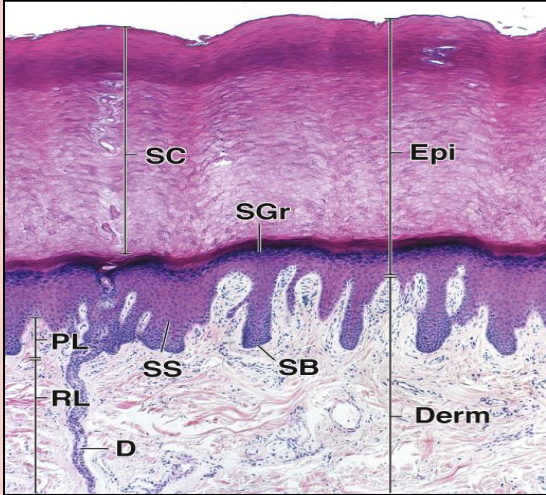
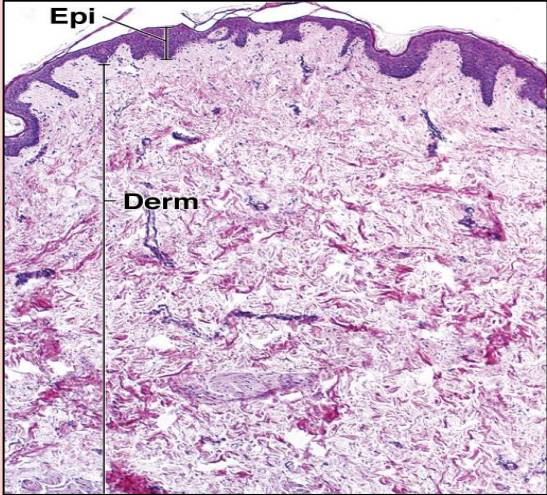
Functions:

- **Immunological role**
 - Encounters, processes and expresses antigens (antigen presenting cell)
 - Migrates to the lymph node to present the antigen to T-lymphocytes
- Express both **MHC I & MHC II** and receptors for IgG
- Involved in **delayed type hypersensitivity reactions**



Thick skin vs Thin skin

The distinction of thick vs think skin is made based on the **thickness of the epidermis**

Thick Skin	Thin Skin
Found only in the palms and soles	Found everywhere else
Significantly thicker epidermis	Thinner epidermis than thick skin
Lack hair follicles	Hair follicles are widespread except for a few locations
Contain a stratum lucidum	Lack a stratum lucidum
	

The Epidermis

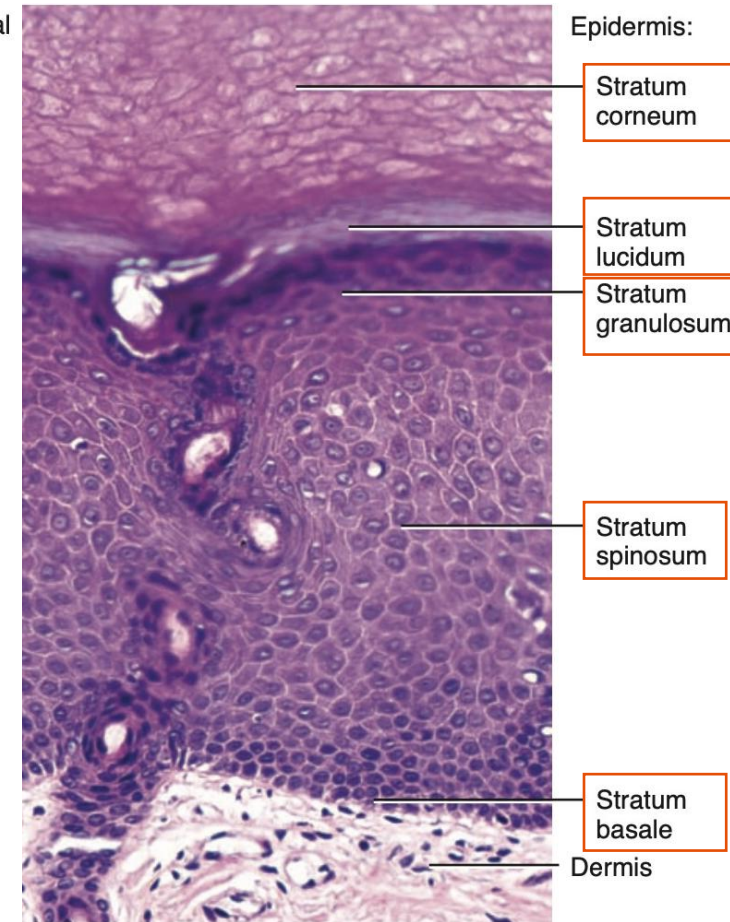
Stratified squamous keratinized epithelium

- Composed of 5 major layers:
 - Stratum Corneum**
 - Dead keratinized cells
 - Stratum Lucidum** – Present in **thick skin only**
 - Cells that are advanced in the keratinization process
 - Stratum Granulosum**
 - Cells that are about to die
 - Stratum Spinosum**
 - Maturing and slowly dying cells
 - Stratum Basale**
 - Stem cells that can differentiate

Come, Let's Get Sun Burnt

Superficial

Deep



Epidermis:

Stratum
corneum

Stratum
lucidum

Stratum
granulosum

Stratum
spinosum

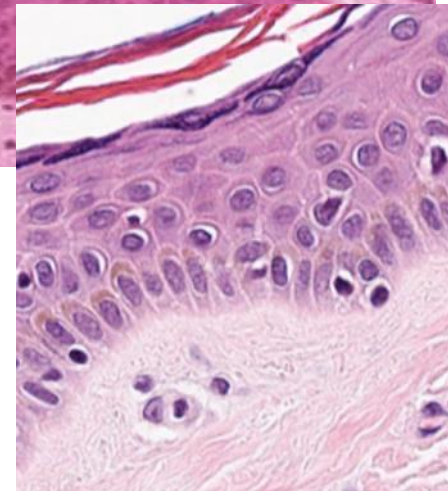
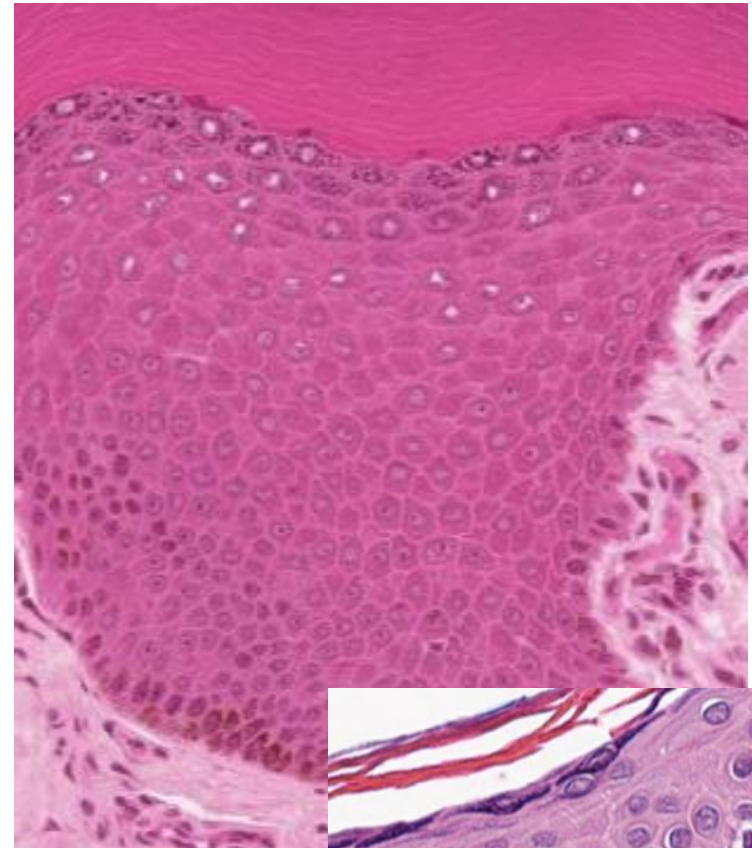
Stratum
basale

Dermis

Stratum Basale

Single layer of cuboidal to low columnar cells

- Mitotically active keratinocytes (stem cells) → **Stratum germinativum**
 - Noticeable **basophilia on H&E staining**
 - Nuclei closely spaced
 - Small cells with scant basophilic staining cytoplasm
 - Synthesis of **keratin filaments**
 - Contains melanin granules
 - Rests on the basal lamina
 - Attached by **hemidesmosomes**
 - Clinical correlate: **Bullous pemphigoid**
 - Extensive cell-to-cell anchoring junctions
 - **Desmosomes**
- **Melanocytes** and **Merkel cells** are also located in this layer



Stratum Spinosum

Several cell layers thick

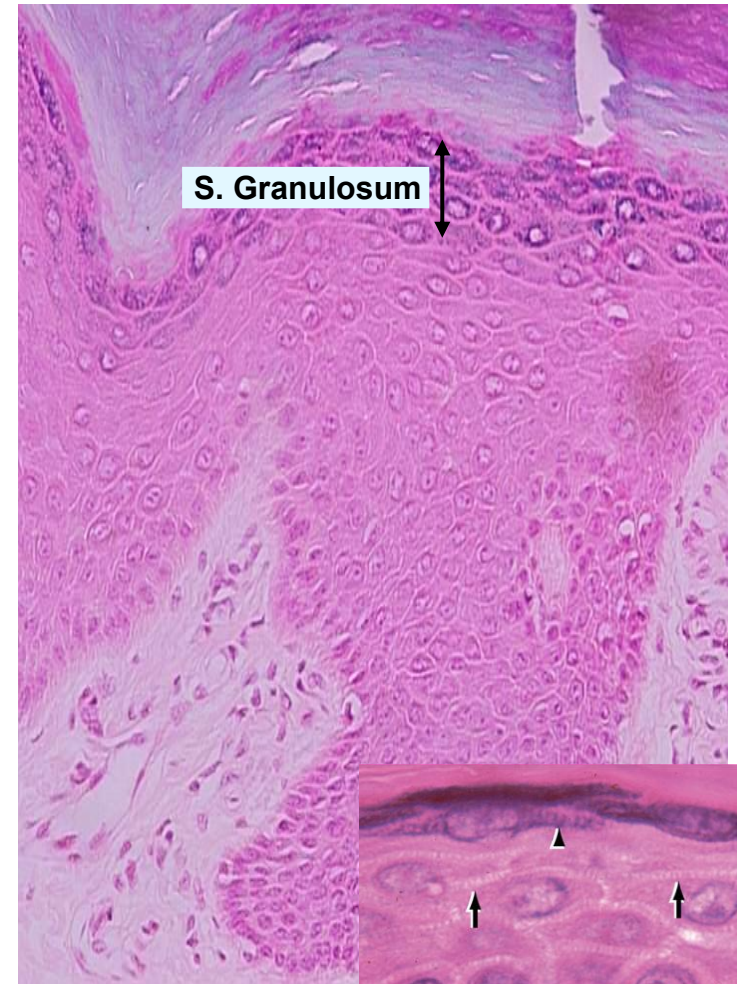
- Larger keratinocytes that gradually change their appearance as they move closer to the surface
 - Cells increase in size and become flattened in a plane parallel to the surface
 - Nuclei become elongated
 - Most noticeable in superficial layers
 - Exhibit numerous cytoplasmic processes or spines → **Prickle cell layer**
 - Processes of neighboring cells are connected to each other via **desmosomes**
 - Appear as a slight thickening and is known as **nodes of Bizzozero**
 - Clinical correlate: **Pemphigus vulgaris**
 - Noticeable **eosinophilia on H&E staining**
 - Synthesis of **tonofibrils** → aggregates of keratin filaments
- **Langerhans cells** are found predominantly in this layer



Stratum Granulosum

Approximately 1-to-3 cell layers thick

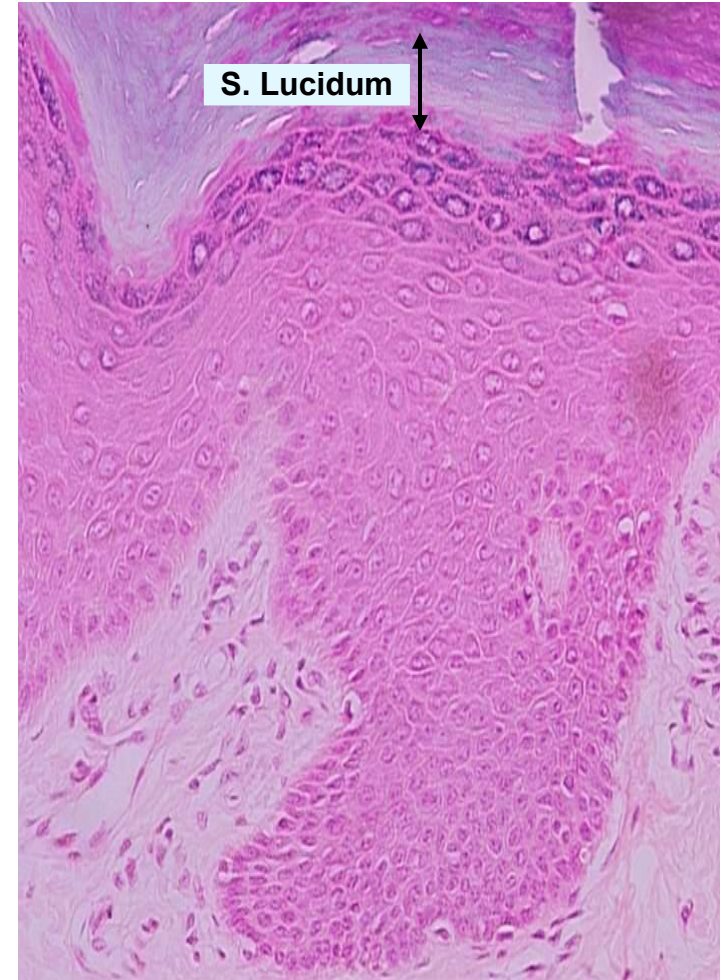
- Most superficial layer of nucleated keratinocytes with conspicuous granules → **granulated appearance**
 - Noticeable **basophilia on H&E staining**
 - **Keratohyalin granules** – irregularly shaped and variably sized granules containing histidine and cystine rich proteins → precursors for **filaggrin**
 - Responsible for aggregating the keratin filaments into tonofibrils
 - Initiation of the keratinization process
 - Conversion of granular cells into cornified cells
 - Synthesis of **Lamellar bodies**
 - Lipid-rich membrane bound granules that aid in formation of the **epidermal water barrier**



Stratum Lucidum

- Only well visualized in **thick skin**
 - Sometimes considered a subdivision of the stratum corneum
- Appears to contain cells in which the keratinization process is well advanced

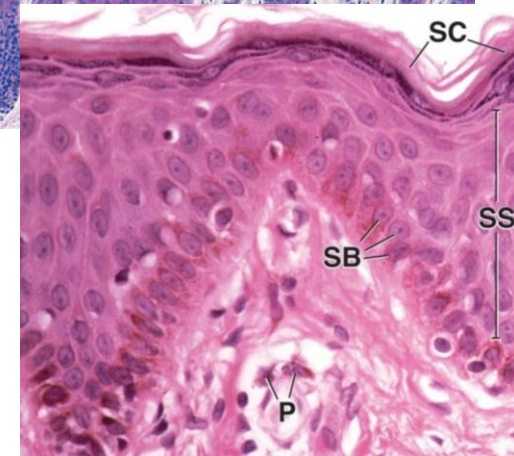
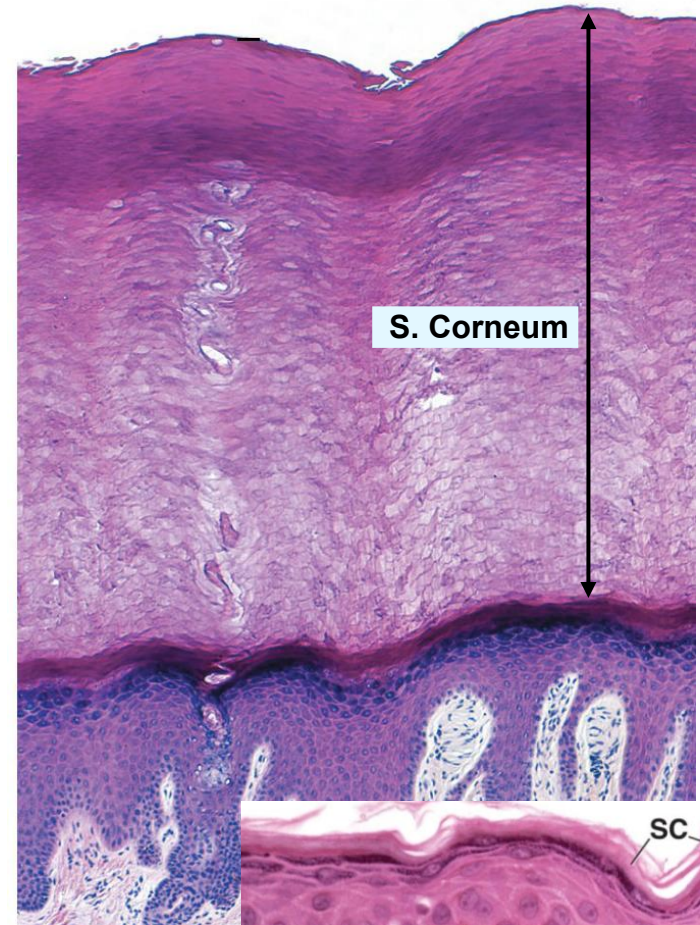
Light microscope → appears as a light highly refractile eosinophilic band



Stratum Corneum

Several cell layers thick

- Variable number of layers of keratinized keratinocytes
 - Point of differentiation between thick and thin skin
 - Layer may thicken under increased amounts of friction
 - Clinical correlate: **Calluses**
 - Anucleate and void of organelles
 - Keratinocytes filled with **soft keratin**
 - In the deeper portions of this layer the thick plasma membranes become coated externally with **lipids**
 - Forms a barrier that is impermeable to water
→ **water barrier**
- Abrupt transition from nucleated stratum granulosum cells



Formation of Epithelial Water Barrier

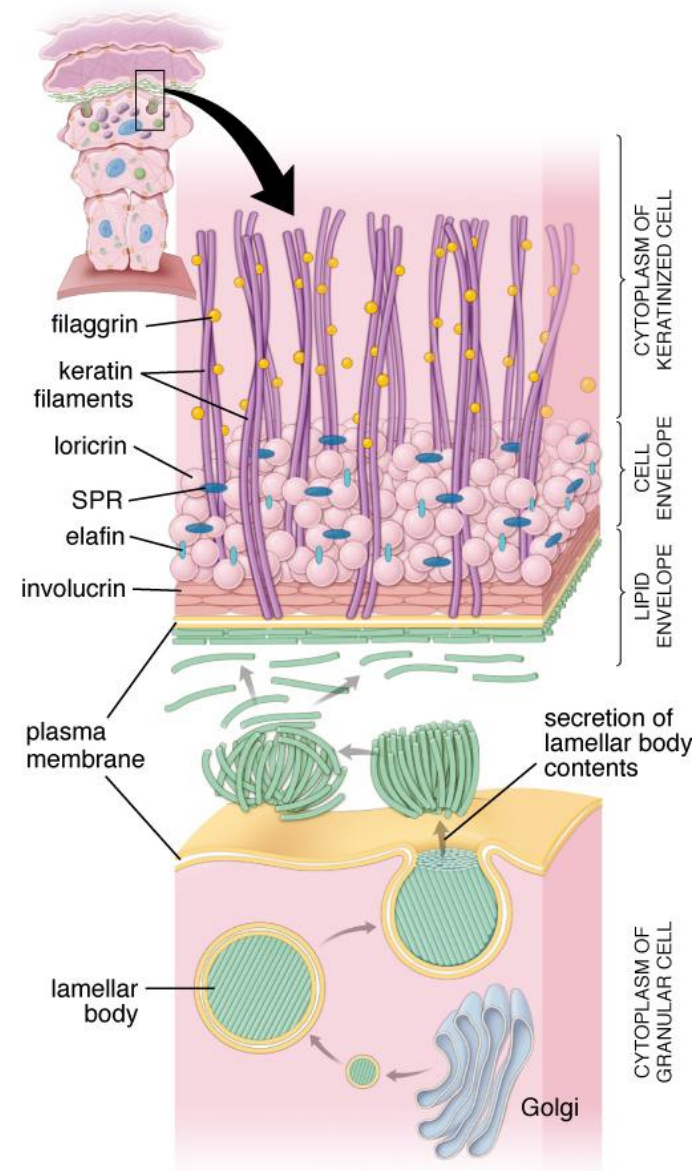
• Cell envelope

- Insoluble protein layer on inner surface of the cell membrane
- Mechanical part of barrier

• Lipid envelope

- Lipid layer on outer surface of the cell membrane
- **Lamellar bodies**
 - Formed by the Golgi apparatus of the spinosum and granular cells
 - Rich in lipids (ceramides, cholesterol) and lipid processing enzymes
 - Secreted by exocytosis into the intercellular spaces between the stratum granulosum and corneum

- Travels with the corneum cells until they are exfoliated



Stop Point 3

A 45-year-old woman comes to the outpatient dermatology clinic with a 3-week history of painful lesions on her skin and painful sores inside her mouth and nose. Vitals are within normal limits. Physical examination reveals multiple blisters on her trunk and several erosions along her buccal and nasal mucosa. Enzyme-linked immunosorbent assay (ELISA) detects IgG autoantibodies against desmoglein 1 and 3 proteins. Which of the following epidermal skin layers is most affected by these autoantibodies?

- A. Stratum corneum**
- B. Stratum lucidum**
- C. Stratum granulosum**
- D. Stratum spinosum**
- E. Stratum basale**
- F. Papillary dermis**

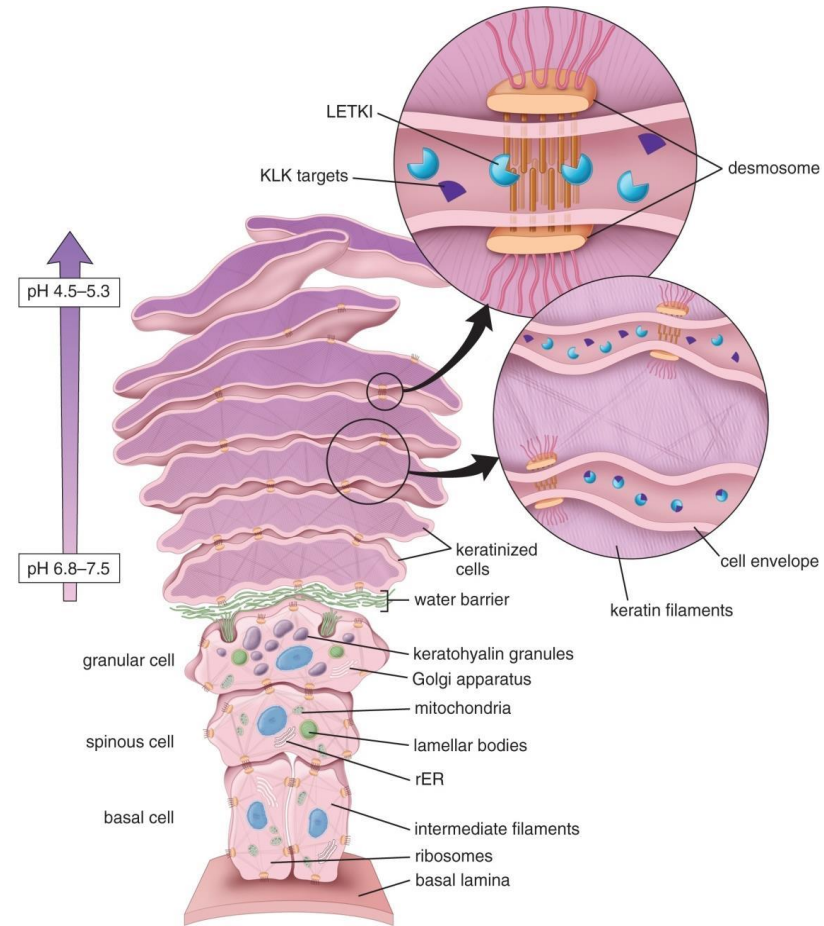
Stop Point 4

Epidermal layer	Characteristics
Stratum Basale	
Stratum Spinosum	
Stratum Granulosum	
Stratum Lucidum	
Stratum Corneum	

Keratinization

A specialized form of **apoptosis** in which cells accumulate **keratin** in their cytoplasm and subsequently slough off

- Keratinocytes in the stratum basale, spinosum and granulosum produce **tonofilaments**
- Within the stratum spinosum and granulosum, **keratohyalin granules** containing **filaggrin** promote aggregation of tonofilaments into **tonofibrils**
 - Granular cells also lose their nuclei, organelles, and cytoplasm as they become cornified
- As keratinocytes ascend to upper layers of the stratum corneum their desmosomes are degraded by **proteolytic enzymes** that become activated due to the change in pH (more acidic)
 - Cells become desquamated



Epithelial Turnover

Normal epidermal turnover time is approximately **47 days**

- Clinical correlate: **Psoriasis**

Days 1-2

Keratinocytes in stratum basale divide mitotically and migrate to stratum spinosum



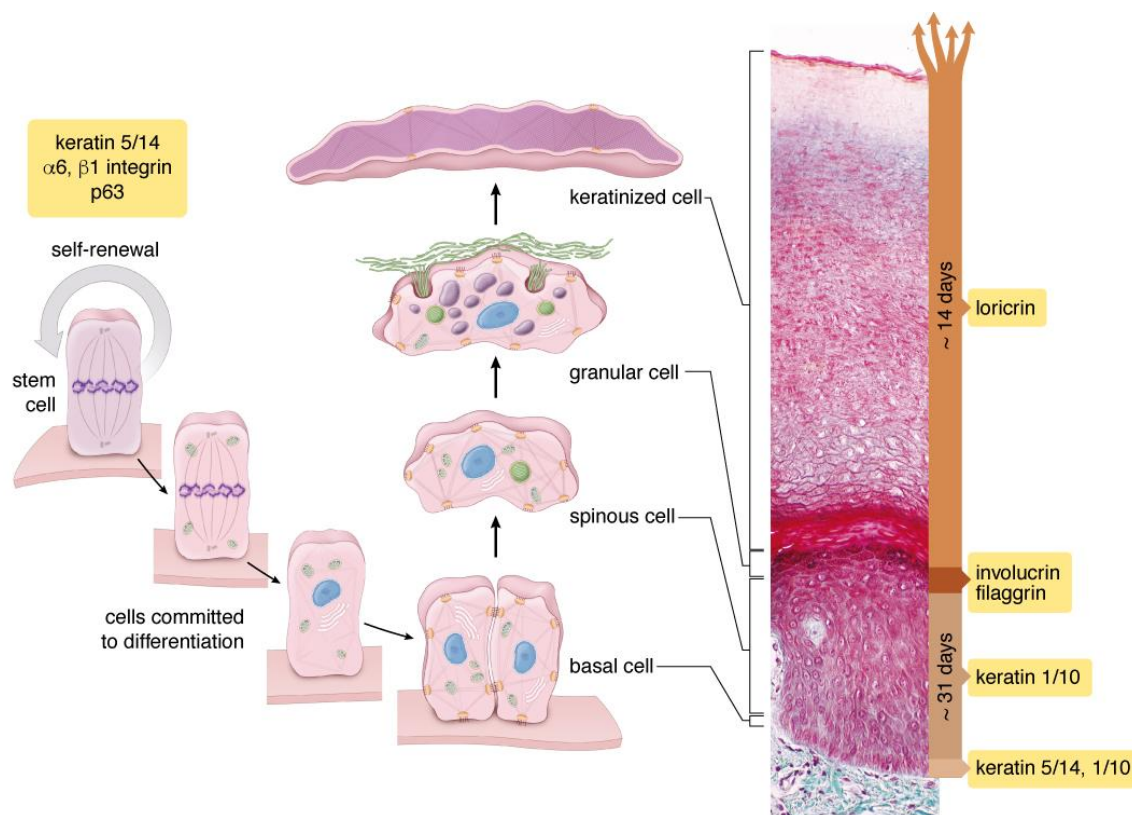
Days 3-31

Duration keratinocytes remain in stratum spinosum
Later move to granulosum



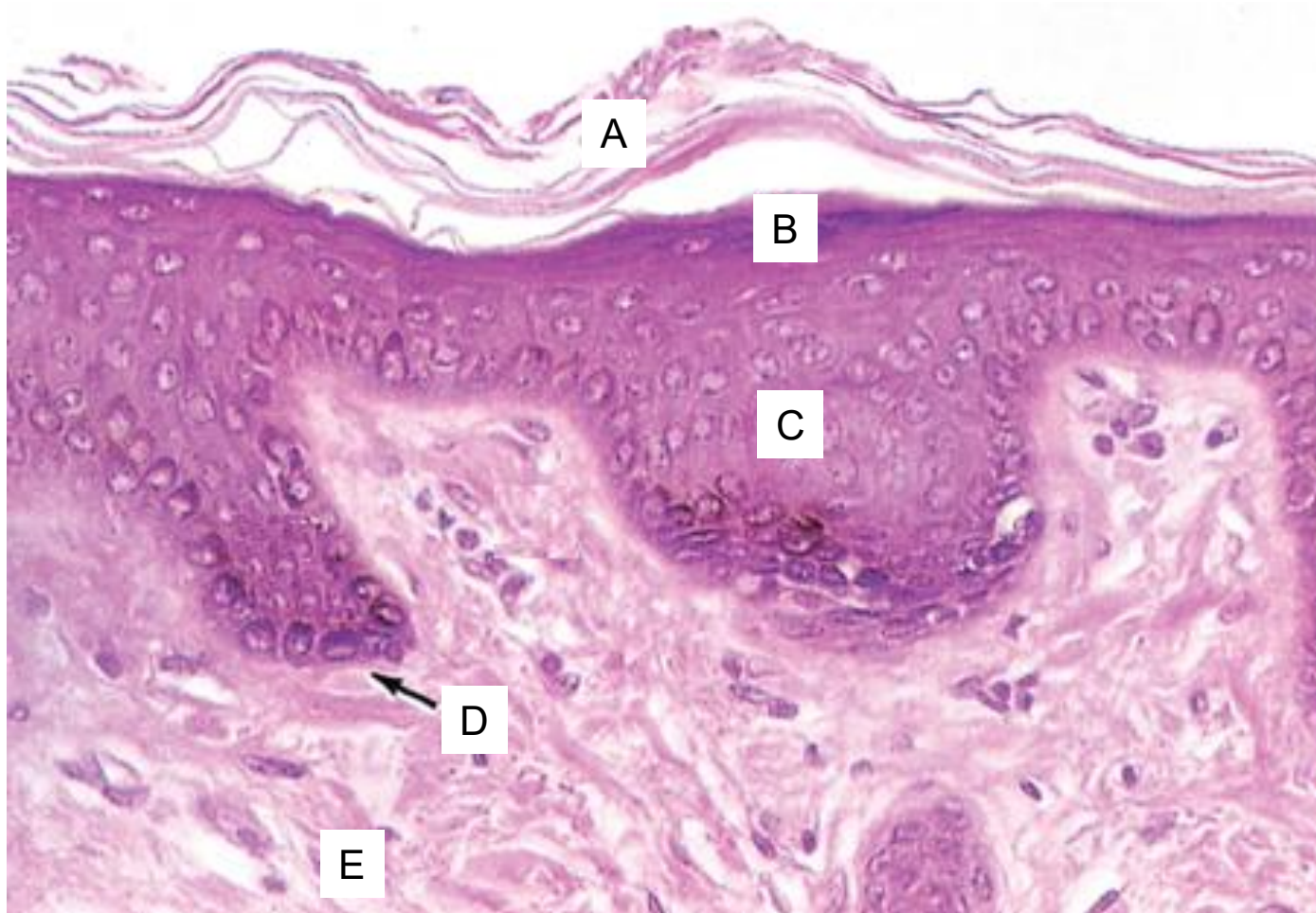
Days 32-46

Keratinocytes become keratinized and enter stratum corneum



Stop Point 6

Identify the following labels:



To Be
Continued...
